

Isometric Exercise & Resting Blood Pressure - Published Abstract (backup copy)

Edwards JJ, Deenmamode AHP, Griffiths M, Arnold O, Cooper NJ, Wiles JD, O'Driscoll JM.
Exercise training and resting blood pressure: a large-scale pairwise and network meta-analysis of randomised controlled trials. *Br J Sports Med.* 2023;57(20):1317-1326.
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Objective: Conduct a comprehensive meta-analysis examining the effects of all relevant exercise training modes on resting blood pressure to establish optimal antihypertensive exercise prescription practices.

Design: Systematic review and network meta-analysis.

Data sources: PubMed (Medline), the Cochrane Library, and Web of Science.

Results: Analysis of 270 randomised controlled trials (15,827 participants) showed significant blood pressure reductions across multiple exercise modes. Isometric exercise training produced the largest reduction at -8.24/-4.00 mm Hg (systolic/diastolic), with combined training at -6.04/-2.54 mm Hg. By effectiveness the ranking was: isometric exercise first, then combined training, dynamic resistance training, aerobic training, and high-intensity interval training (isometric SUCRA 98.3%). Isometric wall squats were the most effective submode for systolic blood pressure.

Conclusions: Various exercise training modes improve resting blood pressure, particularly isometric exercise; the findings are intended to guide future hypertension prevention and treatment guidelines.

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